Highlights

- Sparse feature selection beats added deep capacity for out-of-region transfer.
- A sparse-attention encoder (L-TAE-S) ties TabNet for best transfer, trains faster.
- Field-level aggregation narrows transfer gaps but rarely raises accuracy.
- Model rankings invert between in-region and out-of-region holdout tests.
- Crop models should be tested outside their training region, not only within it.